SANKET · AI

SIF Precursor Detection & Predictive Safety Intelligence for Oil India Limited

SIH26165
Smart India Hackathon 2026
Ministry of Petroleum & Natural Gas

⚠ THE PROBLEM — WHY THIS MATTERS

27 May 2020, Baghjan Well #5: An uncontrolled gas blowout killed 3 people, displaced 8,000+ families, and burned for 172 days — because scattered precursor warning signs (abnormal pressure surges, mud weight loss, BOP weeping seals) were trapped in paper logbooks, never analyzed together.

SANKET · AI ensures this never happens again.

<5s	8	7	10	3	9	24/7
VOICE → SIF SCORE	AI AGENT SWARM	PIPELINE STAGES	OIL ASSET SITES	LANGUAGES (HI/AS/EN)	IOGP LIFE-SAVING RULES	REAL-TIME CLOUD MONITOR

🔍 COMPLETE SAFETY INTELLIGENCE PLATFORM (12 MODULES)

1. **Voice-First Multilingual Reporting** — Workers speak naturally in Hindi, Assamese, or English. OpenAI Whisper transcribes with industrial domain normalization in <3s.
2. **LangGraph 8-Agent Swarm** — Orchestrated multi-agent architecture: Supervisor → Classifier → Entity Extractor → Rule Mapper → Verifier Arbiter → Graph Analyst → Explainer → Action Agent.
3. **Dual-Layer SIF Risk Scoring** — Layer 1: Physics boundary guardrails (pressure/temp). Layer 2: Multilingual E5 neural embeddings. Verifier synthesizes consensus confidence score.
4. **IOGP 9 Life-Saving Rules Mapping** — Auto-aligns observations to global safety standards (Bypassing Safety Controls, Confined Space, Line of Fire, Working at Heights, etc.).
5. **8 Industrial Entity Types** — Automatic extraction of Equipment, Location, Barrier Failure, Energy Source, Toxic Chemical, Personnel, Action, and Severity using specialized NLP.
6. **Live GIS Satellite Heatmap** — 10 real OIL asset coordinates with SIF density heat rings, live wind-directed gas plume dispersion simulation, trunk pipelines, and NASA FIRMS thermal spots.
7. **Safety Knowledge Graph** — Live signal graph connecting equipment, barriers, and failure modes across assets. Detects near-miss multiplier clusters before catastrophic escalation.
8. **Baghjan Disaster Replay Simulation** — 14-day interactive counterfactual replay: demonstrates day-by-day how SANKET would have intercepted the May 2020 blowout 11 days prior.
9. **Centralized Control Room** — Live WebSocket streaming telemetry, high-SIF audio alert horn, active permit tracking, 72-hour trend analytics, and instant supervisor triage dashboard.
10. **Emergency Stop Work Authority (SWA)** — One-click safety interlock: instantly broadcasts SMS/push alarm to all on-site personnel with GPS lockdown zone & emergency muster route.
11. **Automated Shift Handover PDF** — One-click generation of statutory shift handover dossiers containing SIF metrics, top hazard clusters, pending CAPA work orders, and toolpusher sign-off.
12. **FastMCP Tool Registry + CAPA Dispatch** — Model Context Protocol tools expose safety database for AI agents. Action Agent auto-dispatches CAPA corrective actions with SLAs.

🤖 LANGGRAPH 8-AGENT ORCHESTRATION PIPELINE

AGENT 1 — 🎯 Supervisor — Normalize & Lang Detect
→ AGENT 2 — 📊 Classifier — Multilingual E5 SIF Score
→ AGENT 3 — 🔬 Entity Extractor — 8 Industrial Entity Types
→ AGENT 4 — 📋 Rule Mapper — IOGP 9 Life-Saving Rules

▼ DEFENSE-IN-DEPTH CONSENSUS & KNOWLEDGE GRAPH DISPATCH ▼

AGENT 5 · CRITIC — ⚖️ Verifier Arbiter — Physics Guardrails vs Neural
→ AGENT 6 — 🕸️ Graph Analyst — Safety Knowledge Graph
→ AGENT 7 — 💡 Explainer — Plain-English Risk Rationale
→ AGENT 8 — 🦾 Action Agent — CAPA Work Orders & SLA

📊 WHY WE'RE DIFFERENT

Capability	Paper / Manual	Basic Digital	SANKET · AI
Voice-first multilingual hazard input	✗	✗	✓
Multi-agent AI orchestration (8 agents)	✗	✗	✓
Dual-layer SIF scoring + adversarial verification	✗	✗	✓
Real-time GIS heatmap + gas plume simulation	✗	✗	✓
Knowledge Graph + near-miss cluster detection	✗	✗	✓
Baghjan disaster counterfactual simulation	✗	✗	✓
Report-to-dashboard latency	24–72 hrs	1–4 hrs	**< 5 sec**

🛠 TECHNOLOGY STACK

Python / FastAPI · Next.js 15 · LangGraph (Multi-Agent) · OpenAI Whisper · GPT-4o-mini · Multilingual E5 Embeddings · Supabase (PostgreSQL) · WebSocket (Real-time) · Leaflet.js (GIS/Satellite) · FastMCP (Tool Registry) · TailwindCSS · Docker · Railway (Cloud) · HuggingFace · Pydantic · SQLAlchemy

Regulatory Compliance: OISD-GDN-206 (Safety Mgmt) · IS 15656 (Hazard ID) · IOGP 9 Life-Saving Rules · PNGRB Upstream Safety · OIL HSE Policy · OSHA PSM 1910.119

Team SANKET · Smart India Hackathon 2026 · Problem Statement: **SIH26165**
Developed for **Oil India Limited (OIL)** · Ministry of Petroleum & Natural Gas

"Every precursor detected is a disaster prevented — Protecting lives through predictive intelligence."

LIVE CLOUD DEMO
`oil-sanket.up.railway.app`
Scan with phone camera for instant demo
Worker Mobile Terminal: /worker